Product Manufacture Exploration

Predicting Failures of Super Soaker

Overview

Purpose: Predict the product failure of Super Soaker from given data

Most influential predictor variables:

loading > measurement_17 > measurement_8 > measurement_2

Prediction models executed include:

- Logistic regression
- Classification tree
- Random forest
- XGBoost
- Neural Network

Train/ test split ratio: 70/30

Balances model learning while reserving a substantial portion for evaluation

Data Limitations and Model Mitigations

Limitation: Missingness

Model Mitigation: Imputation method applied to handle missingness while maintaining data integrity

Limitation: Outliers

Model Mitigation: Evaluated and filtered outliers to reduce influence

Limitation: Class Imbalance/ Skew

Model Mitigation: Used performance metrics of accuracy, sensitivity, and specificity to limit bias

Model Descriptions

Logistic Regression

Estimates the probability of an outcome using weighted predictor variables

Classification Tree

Splits data into decision rules, forming a tree structure that predicts outcomes based on sequential conditions

Random Forest

An ensemble of many decision trees that “vote” on the final prediction

XGBoost

Builds trees sequentially with each correctly the errors of the previous one

Neural Network

Composed of interconnected layers of nodes that learn complex patterns through weighted connections and activation functions

Model	Predictors	train_set			test_set		
		Accuracy	Sensitivity	Specificity	Accuracy	Sensitivity	Specificity
Classification Tree	all	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
	loading	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
	loading + measurement_17	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
	loading + measurement_17 + measurement_2	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
Random Forest	all	0.7953	0.0323	1.0000	0.7888	0.0024	0.9998
	loading	0.8507	0.3844	0.9767	0.7274	0.1327	0.8880
	loading + measurement_17	0.9162	0.6082	0.9993	0.7707	0.0502	0.9653
	loading + measurement_17 + measurement_8	0.9283	0.6629	0.9999	0.7802	0.0159	0.9866
	all	0.7953	0.0323	1.0000	0.7888	0.0024	0.9998
	loading	0.8131	0.1432	0.9929	0.7668	0.0495	0.9593
	loading + measurement_17	0.8052	0.0812	0.9995	0.7851	0.0110	0.9928
	loading + measurement_17 + measurement_8	0.8004	0.0573	0.9997	0.7858	0.0031	0.9958
XGBoost	all	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
	loading	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
	loading + measurement_17	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
	loading + measurement_17 + measurement_8	0.7885	0.0000	1.0000	0.7885	0.0000	1.0000
Neural Network	all	0.7886	0.0007	1.0000	0.7885	0.0000	1.0000
	loading	0.7886	0.0020	0.9996	0.7880	0.0008	0.9992

Model Results/ Performance Based on Accuracy

Model Interpretations and Next Steps

- Random Forests were the only models achieving reasonable accuracy, reflecting their strength in capturing nonlinear relationships and interactions among predictors
- Large gaps between training and testing set performance indicated overfitting, suggesting that models learned “noise” in the training data rather than generalized patterns.
- All models struggled with the imbalanced failure outcome, resulting in low sensitivity and a tendency to predict the majority class.
- No model demonstrated minority class detection, limiting practical use for identifying true failure cases.
- Performance patterns highlight the need for enhanced imbalance handling, such as class weighting, resampling, or alternative algorithms.

Project Summary



BUILT AND EVALUATED MULTIPLE PREDICTIVE MODELS TO CLASSIFY FAILURE OUTCOMES USING A 70/30 TRAIN-TEST SPLIT



ADDRESSED KEY DATA CHALLENGES – MISSINGNESS, OUTLIERS, AND A HIGHLY IMBALANCED FAILURE VARIABLE (79% NONFAILURE, 21% FAILURE)



RANDOM FOREST MODELS DELIVERED THE STRONGEST OVERALL ACCURACY BUT SHOWED SUBSTANTIAL TRAIN-TEST SET GAPS, INDICATING OVERFITTING



ALL MODELS STRUGGLED TO DETECT MINORITY FAILURE CASES, LIMITING SENSITIVITY AND PRACTICAL UTILITY



FINDINGS HIGHLIGHT THE NEED FOR IMPROVED IMBALANCE HANDLING TO ACHIEVE RELIABLE MINORITY CLASS PREDICTION.